IN THE HIGH COURT OF JUSTICE
BUSINESS AND PROPERTY COURTS OF ENGLAND AND WALES
TECHNOLOGY AND CONSTRUCTION COURT (KBD)

Claim No. HT-2025-000418

HARROW PRECISION ENGINEERING LIMITED *(Claimant)* v BREVDALE CASTINGS LIMITED *(Defendant)*

EXPERT REPORT OF DR FAISAL IQBAL

CEng FIMMM — Metallurgy and Foundry Engineering

Prepared on the instructions of the Claimant · CPR Part 35

Dated 24 January 2026

1. Introduction and instructions

I am Dr Faisal Iqbal, Principal Metallurgist at Wraysbury Materials Laboratory, a UKAS-accredited testing laboratory (accreditation no. 4471). I hold a PhD in physical metallurgy and am a Chartered Engineer and Fellow of the Institute of Materials, Minerals and Mining, with twenty-six years' experience in casting integrity and failure analysis.

I have been instructed by Whitfield Carter LLP, solicitors for the Claimant, to give my independent opinion on the metallurgical quality of ductile-iron pump housing castings (part numbers HPE-PH-220 and HPE-PH-340) supplied by the Defendant to the Claimant between 4 March and 21 June 2024, and on the cause of the field failures reported in service. I had previously reported to the Claimant on a non-litigation basis on 26 September 2024; this report is my CPR Part 35 report and supersedes that earlier document.

2. The questions I was asked to address

1. Whether the castings conformed to grade EN-GJS-400-15 (BS EN 1563:2018) and to acceptance level 2 of BS EN 12681 in the pressure-containing regions.
2. The nature, cause and origin of any porosity found, and whether it was present as-cast or introduced by the Claimant's machining or proof-testing.
3. Whether the sample examined was representative, and the proportion of the population properly regarded as non-conforming.
4. Whether the field failures were consistent with the porosity found.

3. Documents and materials examined

I examined 400 castings selected at random by my laboratory across all eighteen deliveries in the relevant period; the Defendant's foundry inspection certificates; specification HPE-MS-014 (Rev 4); the relevant drawings; and eleven failed housings returned from the field. Testing comprised digital radiography to BS EN 12681, dye-penetrant inspection to BS EN 1371-1, spectrographic analysis, tensile testing to BS EN 1563, and metallographic sectioning of representative defects.

4. Findings on conformity

Radiographic examination of the 400-casting sample disclosed sub-surface gas porosity and shrinkage cavities in the pressure-containing wall of the housing. Applying acceptance level 2 of BS EN 12681 to the pressure-containing regions defined on the drawing, 312 of the 400 castings (78.0%) failed. The defects were concentrated in the cored waterway and the boss transition, both of which are pressure-containing. I do not accept the Defendant's contention that the indications lie only in non-critical flange areas; the metallographic sections (Appendix C) show porosity directly within the 16 bar pressure boundary.

Tensile specimens machined from affected regions returned elongation values between 6% and 11%, below the 15% minimum for grade EN-GJS-400-15. The material in the affected regions therefore did not meet the specified grade.

5. Cause and origin of the porosity

The morphology of the defects — rounded gas porosity associated with interdendritic shrinkage — is characteristic of inadequate melt de-gassing combined with an unfavourable gating and feeding arrangement. The porosity is metallurgical in origin and was present in the casting as solidified. It is not machining damage: machining produces surface tearing or smearing, not the internal rounded porosity seen here. Nor is it test-induced: a proof test does not create porosity; at most it reveals a casting already weakened by it.

I have reviewed the Claimant's CNC machining sequence and its 24 bar proof test (1.5 times the 16 bar working pressure, which is the conventional proof factor). Neither could have caused the defects I observed. The defects originate in the foundry process operated by the Defendant.

6. Representativeness and extrapolation

The 400-casting sample was drawn at random by my laboratory across all relevant deliveries, not from a single order. A failure rate of 78.0% in a random sample of that size gives a 95% confidence interval of approximately 73.6% to 81.9%. Applied to the population of 1,840 castings, the central estimate is 1,435 non-conforming castings, with a lower 95% bound of approximately 1,354. The Claimant's pleaded figure of 'not fewer than 1,430' is, if anything, conservative and is well supported by the data.

7. The field failures

All eleven failed housings I examined exhibited leakage paths running through sub-surface porosity in the pressure-containing wall. The failures are fully consistent with, and in my opinion caused by, the casting porosity. They are not consistent with assembly error, gasket failure or over-pressurisation, each of which I considered and excluded.

8. Summary of opinions

1. A substantial majority of the castings supplied between March and June 2024 did not conform to grade EN-GJS-400-15 or to acceptance level 2 of BS EN 12681 in the pressure-containing regions.
2. The porosity was metallurgical, present as-cast, and caused by the Defendant's de-gassing and gating practice. It was not caused by the Claimant's machining or proof-testing.
3. On the sampling, not fewer than approximately 1,354 and most probably about 1,435 of the 1,840 castings are non-conforming.
4. The field failures were caused by the casting porosity.

Declaration

I understand that my overriding duty is to the Court and I have complied with that duty. I have set out in this report those matters within my own knowledge and those on which I have relied. I confirm that I have made clear which facts and matters are within my own knowledge and which are not, and that the opinions I have expressed represent my true and complete professional opinion. I confirm that I am aware of and have complied with the requirements of CPR Part 35, Practice Direction 35 and the Guidance for the Instruction of Experts in Civil Claims.

Statement of truth: I confirm that I have made clear which facts and matters referred to in this report are within my own knowledge and which are not. Those that are within my own knowledge I confirm to be true. The opinions I have expressed represent my true and complete professional opinions on the matters to which they refer. I understand that proceedings for contempt of court may be brought against anyone who makes, or causes to be made, a false statement in a document verified by a statement of truth without an honest belief in its truth.

Signed: Dr Faisal Iqbal CEng FIMMM

Wraysbury Materials Laboratory · 24 January 2026